APPSC GROUP 1 / GROUP 2 — GEOGRAPHY

Volume 2: Physical Geography of India and Andhra Pradesh

Location · Physiography (Himalaya, Plains, Plateau, Desert, Coasts, Islands) · Drainage · Climate and Monsoon · Soils · Natural Vegetation and Wildlife · Natural Hazards and Disaster Management

READ THIS FIRST

How to use: same pattern as Volume 1: **Blue** = simple explanation · **Orange** = Andhra Pradesh / Kurnool link · **Red** = exam traps · **Green** = 2-minute revision · **Purple** = self-check. Use the **maps** while reading and later try to redraw them from memory.
Study plan (6 days): Day 1 Ch 1-2 · Day 2 Ch 3 (Himalayas and plains) · Day 3 Ch 4 (rivers) · Day 4 Ch 5 (climate) · Day 5 Ch 6-7 (soil, vegetation) · Day 6 Ch 8 (hazards) + final test.
Syllabus match (Group 1 and 2): 'India and Andhra Pradesh: Major physiographic features, Climate, Drainage System, Soils and Vegetation, Natural Hazards and Disasters and their Management'. This volume covers all of it.
Note: maps are schematic (not to exact scale). Figures marked (verify) can change; recheck them in the latest current affairs notes.

CHAPTER 1 — INDIA: LOCATION, SIZE AND NEIGHBOURS

Where is India on the globe?

IN SIMPLE WORDS (what / how / why)

Latitude: 8°4′N (Kanyakumari, mainland) to 37°6′N. **Longitude:** 68°7′E to 97°25′E. The **Tropic of Cancer (23.5°N)** divides India almost into two equal halves: the south is **tropical**, the north is **sub-tropical**.
Size: about **3.28 million sq km**, the **7th largest country** in area (Russia, Canada, China, USA, Brazil, Australia are bigger), about **2.4%** of the world's land but about **17.5%** of its people.
Why is India's location important? (1) Central position in Asia between the East and West. (2) The **Indian Ocean** gives sea routes. (3) The **Himalaya** keeps cold Siberian winds out and gives snow-fed rivers. (4) The **Deccan** juts into the ocean, so the Indian Ocean is named after India.
Southernmost point of India = Indira Point (Great Nicobar) at about 6°45′N (mainland: Kanyakumari). **Standard time meridian = 82°30′E** (near Mirzapur, UP).

Neighbours (7 land + 2 sea)

Type	Countries	Remember
Land (7)	Pakistan, Afghanistan, China, Nepal, Bhutan, Myanmar, Bangladesh	**Longest border: Bangladesh** (about 4,100 km). Afghanistan border is the shortest (the Wakhan side) and is only through PoK as per India's position.
Sea (2)	Sri Lanka, Maldives	**Palk Strait** (India–Sri Lanka); **Gulf of Mannar**; **Ten Degree Channel** (Andaman | Nicobar); **Eight Degree Channel** (Minicoy | Maldives).
Border lines	Radcliffe (India–Pakistan, 1947); Durand (Afghanistan–Pakistan); McMahon (India–China, east)	LAC = Line of Actual Control with China.

Coastline: about **7,516 km** including islands (mainland about 6,100 km). **9 coastal states:** Gujarat (longest), Maharashtra, Goa, Karnataka, Kerala, Tamil Nadu, **Andhra Pradesh**, Odisha, West Bengal. Coastal UTs: Puducherry, Lakshadweep, Andaman & Nicobar, Dadra & Nagar Haveli and Daman & Diu.
Land boundary: about 15,100 km, shared by 17 states/UTs (figures verify).

ANDHRA PRADESH / KURNOOL CONNECTION

Andhra Pradesh coast: about 1,053 km (among the longest of Indian states; Gujarat is first; verify rank). AP lies between about **12°41′N and 19°55′N** and **76°45′E and 84°46′E** (approx.), entirely in the **tropical zone, south of the Tropic of Cancer**.
Kurnool is at about **15.8°N, 78°E**: inland, on the **Deccan plateau**, in the **rain shadow** zone.

2-MINUTE REVISION

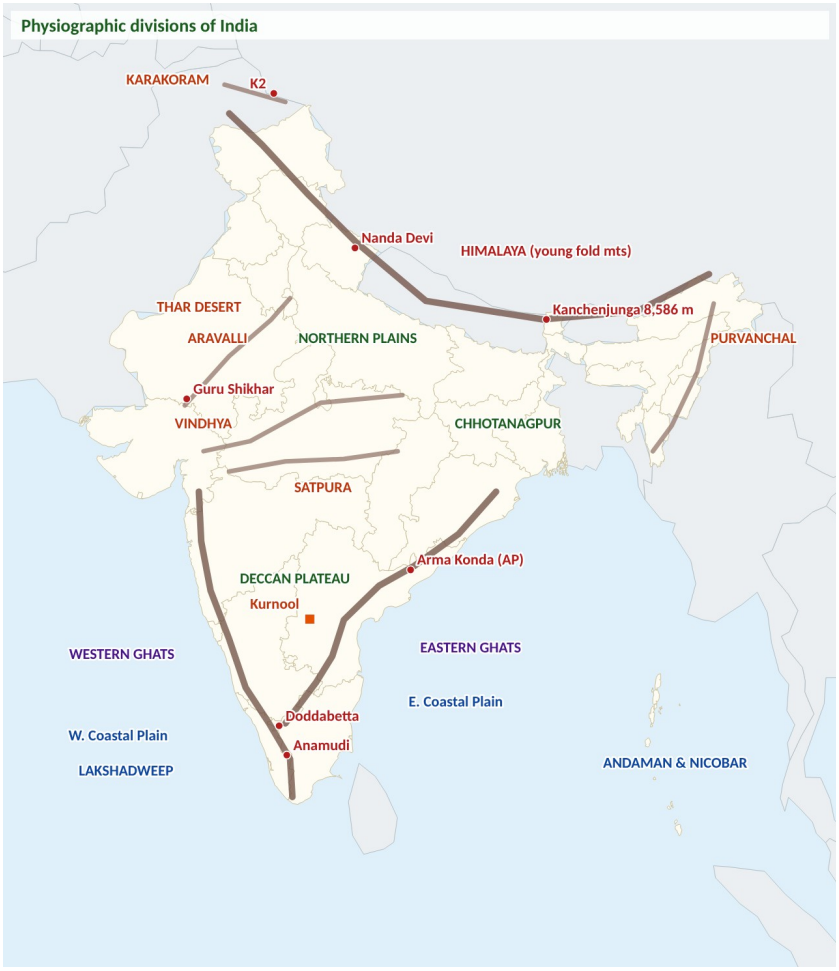
Lat 8°4′N-37°6′N · Long 68°7′E-97°25′E · Tropic of Cancer halves India · 7th largest · 7 land neighbours · Indira Point south · IST 82.5°E · coastline 7,516 km · 9 coastal states · AP coast about 1,053 km.

CHECK YOURSELF

Q1. Longest land border with: **Bangladesh** · Q2. Channel between Minicoy and Maldives: **Eight Degree Channel** · Q3. State with longest coastline: **Gujarat**

CHAPTER 2 — PHYSIOGRAPHIC DIVISIONS OF INDIA

Six big landform regions: how and why they formed



Map 2.1 Physiographic divisions (schematic)

IN SIMPLE WORDS (what / how / why)

Physiography = study of the natural features of land (mountains, plains, plateaus). India has **six** divisions: **(1) Northern Mountains (Himalaya), (2) Northern Plains, (3) Peninsular Plateau, (4) Indian (Thar) Desert, (5) Coastal Plains, (6) Islands.** **Why so different?** Remember Volume 1: India was part of **Gondwanaland** and is the **oldest, hardest** block (the plateau). It then drifted north and **collided** with Asia, **pushing up the Himalaya** (young). The rivers from the Himalaya then **filled the gap** in between with silt and made the plains. So: **old plateau + young mountains + plains made by rivers.**

2.1 The Himalaya (Northern Mountains)

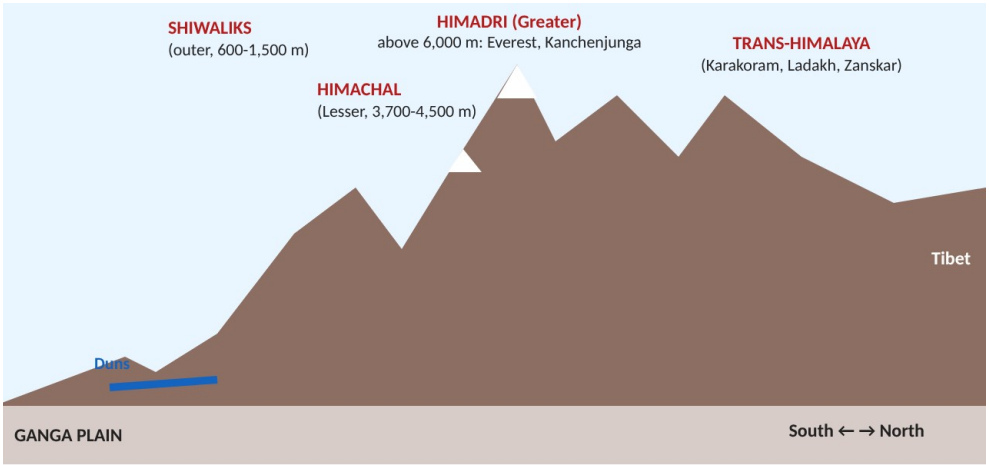


Fig 2.2 Cross-section of the Himalaya (south to north)

IN SIMPLE WORDS (what / how / why)

Young fold mountains, about **2,400 km long**; width 400 km (Kashmir) to 150 km (Arunachal). They are still rising, so earthquakes

and landslides are common. They are in **three parallel ranges** (south to north):
1. Shiwaliks (Outer Himalaya) — lowest, 600-1,500 m; made of loose river sediments; **Duns** (valleys, e.g. Dehradun) between them and the Lesser Himalaya. **2. Himachal / Lesser Himalaya** — 3,700-4,500 m; **Pir Panjal** (longest range), Dhauladhar, Mussoorie; **valleys of Kashmir, Kullu, Kangra**; **hill stations** Shimla, Mussoorie, Nainital, Darjeeling. **3. Himadri / Greater Himalaya** — highest, average above 6,000 m, always snow-covered: **Kanchenjunga (8,586 m; India's highest peak, 3rd in the world)**, **Nanda Devi (7,816 m; highest peak entirely in India)**; Mt Everest is in Nepal. Beyond it is the **Trans-Himalaya**: **Karakoram (K2 = Godwin-Austen, 8,611 m, world's 2nd highest; in PoK/Gilgit, claimed by India)**, Ladakh and Zanskar ranges.

Pass	Connects	Note
Zoji La	Srinagar – Leh	Kashmir to Ladakh
Banihal	Jammu – Srinagar	Jawahar tunnel
Khardung La	Leh – Nubra	One of the highest motorable roads
Rohtang (Atal Tunnel)	Manali – Lahaul	Atal tunnel (2020) under it
Shipki La	Himachal – Tibet	Sutlej enters India
Nathu La, Jelep La	Sikkim – Tibet	Border trade with China
Lipulekh	Uttarakhand – Tibet	Kailash Mansarovar route
Bomdi La	Arunachal – Tibet	

Glaciers: Siachen (longest outside polar regions, Karakoram), **Gangotri** (source of the Ganga's Bhagirathi), **Yamunotri**, **Zemu** (Sikkim).
Eastern hills (Purvanchal): Patkai, Naga, Manipur, Mizo (Lushai) hills, and **Garo-Khasi-Jaintia** (Meghalaya plateau, the wettest region). Rivers: Barak.
Regional divisions by rivers: Punjab Himalaya (Indus–Sutlej), Kumaon (Sutlej–Kali), Nepal (Kali–Tista), Assam (Tista–Dihang).

2.2 The Northern Plains

IN SIMPLE WORDS (what / how / why)

What: a vast flat land of **alluvium** (silt, sand, clay) about **2,400 km long and 240-320 km wide**, formed by the **Indus, Ganga and Brahmaputra** carrying silt from the Himalaya for millions of years (the sea-like trough was slowly filled).
Four belts, from the mountain foot to the south: Bhabar (pebbles at the foothill, streams disappear underground) → **Terai** (the water comes out again; swampy, forests, now farms) → **Bhangar** (older alluvium, higher, has **kankar** lime nodules) → **Khadar** (newer, flood-fed alluvium, very fertile).
Parts: Punjab plains (west, five rivers), **Ganga plains** (largest; Haryana to Bengal), **Brahmaputra plains** (Assam). **Doab** = land between two rivers.
Importance: most fertile region, densest population; wheat, rice, sugarcane, jute. Often flooded.

2.3 The Peninsular Plateau

IN SIMPLE WORDS (what / how / why)

What: a **triangular, very old, hard** block of crystalline rock (part of Gondwana). Its edges are the **Aravalli** (NW), **Vindhya and Satpura** (north), **Western Ghats** (west) and **Eastern Ghats** (east).
Two parts: Central Highlands (north of Narmada: **Malwa**, Bundelkhand, Baghelkhand, **Chhotanagpur** plateau rich in minerals) and **Deccan Plateau** (south of Narmada; tilted **west to east**, so most big rivers flow to the **Bay of Bengal**).
Deccan Traps: about 65 million years ago huge **basalt lava** flows covered much of the plateau; weathering made the **black cotton soil (regur)**.
Aravalli = oldest fold mountains of India (worn down); highest peak **Guru Shikhar (1,722 m, Mt Abu)**. Vindhya and Satpura are block/residual hills with the **Narmada–Tapti rift valleys** between them.

	Western Ghats (Sahyadri)	Eastern Ghats
Shape	Continuous, a steep wall facing the Arabian Sea	Broken, discontinuous; cut by the Godavari, Krishna, Mahanadi and Kaveri
Height	Higher: **Anamudi 2,695 m** (Anaimalai) — highest peak of South India; Doddabetta 2,637 m (Nilgiris)	Lower: **Jindhagada / Arma Konda about 1,680 m (AP)**; Mahendragiri (Odisha) 1,501 m
Rain	Heavy on the west (windward)	Gets less SW monsoon; gets NE monsoon rain
Gaps	**Thal, Bhor, Pal Ghat** passes	Few passes, many river gaps
Meet	They meet in the **Nilgiri hills**	
Other	UNESCO World Heritage Site, biodiversity hotspot	Nallamala, Seshachalam, Velikonda, Palkonda, Javadi, Shevaroy

2.4 Thar Desert, Coastal Plains and Islands

Thar (Marusthali): west of the Aravalli, Rajasthan; rainfall below 25 cm; **sand dunes, barchans (crescent dunes)**; **Luni** (inland river, flows to Rann of Kutch); salt lakes **Sambhar** (largest inland salt lake).
Western Coastal Plain (narrow; Arabian Sea): **Konkan** (Mumbai–Goa), **Kannad / Karnataka coast**, **Malabar** (Kerala: **backwaters / kayals / lagoons**, e.g. **Vembanad**). Estuaries, not deltas.
Eastern Coastal Plain (wide; Bay of Bengal): **Utkal** (Odisha; Chilika lake), **Northern Circars** (Mahanadi to Krishna: north and central AP), **Coromandel** (Krishna to Kanyakumari: south AP and Tamil Nadu). Big **deltas** (Mahanadi, Godavari, Krishna, Kaveri).
Islands: Andaman (north) and **Nicobar** (south), separated by the **Ten Degree Channel**; **Saddle Peak (737 m)** highest; **Barren Island = India's only active volcano**; **Narcondam = dormant**; **Indira Point** in Great Nicobar. **Lakshadweep** (coral atolls, capital Kavaratti; **Minicoy** south). Coral islands vs volcanic islands: Andaman-Nicobar are the tops of a submerged mountain chain (an extension of the Arakan Yoma).

ANDHRA PRADESH / KURNOOL CONNECTION

AP's landforms in one line: Eastern Ghats on the north-east and centre (**Nallamala** between the Krishna and Penna; **Seshachalam** near Tirupati; **Velikonda** and **Palkonda**), the **Rayalaseema plateau** (Anantapur, Kurnool, Kadapa, Chittoor) with low hills and rocky plains, and a **wide coastal plain** with the **Godavari and Krishna deltas**.
Highest point: Jindhagada / Arma Konda about 1,680 m (Alluri Sitarama Raju district); **Araku valley** (Visakhapatnam hills) is a tourist area and a coffee region; **Horsley Hills** (Annamayya, hill station; verify); **Papi Kondalu** gorge on the Godavari.
Kurnool district: plateau at about 200-400 m, the **Nallamala and Erramala hills**, black-soil plains along the **Tungabhadra** and **Handri** rivers; the **Krishna** forms its eastern boundary at **Srisailam**.

EXAM TRAPS

Himalaya = **young fold**; Aravalli = **old fold (oldest)**; Eastern Ghats are NOT a continuous range.
Kanchenjunga is India's highest peak; **Nanda Devi is the highest entirely inside India**; K2 is in PoK.
Western Ghats are HIGHER than the Eastern Ghats.
Bhabar has pebbles (streams vanish); **Terai** is swampy.

2-MINUTE REVISION

Six divisions. Himalaya: Shiwalik < Himachal < Himadri; Trans: Karakoram, Ladakh, Zanskar.
Plains: Bhabar → Terai → Bhangar → Khadar.
Plateau: Central Highlands + Deccan; Deccan Traps = black soil.
Western Ghats: Anamudi. Eastern Ghats: Arma Konda (AP). Thar: Luni, Sambhar.
Coasts: Konkan, Kannad, Malabar | Utkal, Northern Circars, Coromandel.

CHECK YOURSELF

Q1. Highest peak of the Eastern Ghats: **Jindhagada (Arma Konda), AP** · Q2. Only active volcano of India: **Barren Island** · Q3. Northern Circars coast lies between: **Mahanadi and Krishna**

CHAPTER 3 — DRAINAGE: RIVERS, LAKES AND PROJECTS

Where does the water come from and where does it go?

IN SIMPLE WORDS (what / how / why)

Drainage = the system of rivers draining a region. **Watershed (divide)** = the high land that separates two river basins. **River basin** = the whole area a river and its branches drain.
Two big groups: (1) Himalayan rivers — born in snowy mountains, **perennial** (water all year: snow melt + rain), deep gorges in the mountains, **delta** in the plains, often change course and **flood**. **(2) Peninsular rivers** — born in the plateau, **seasonal** (rain-fed, shrink in summer), **older and stable**, shallow valleys, fixed courses.
Why do Peninsular rivers flow east? The Deccan slopes **west to east**, so most rivers flow east into the Bay of Bengal and make **deltas**. Only **Narmada and Tapti** flow west because they run in **rift valleys** (sunken blocks), so they form **estuaries**.

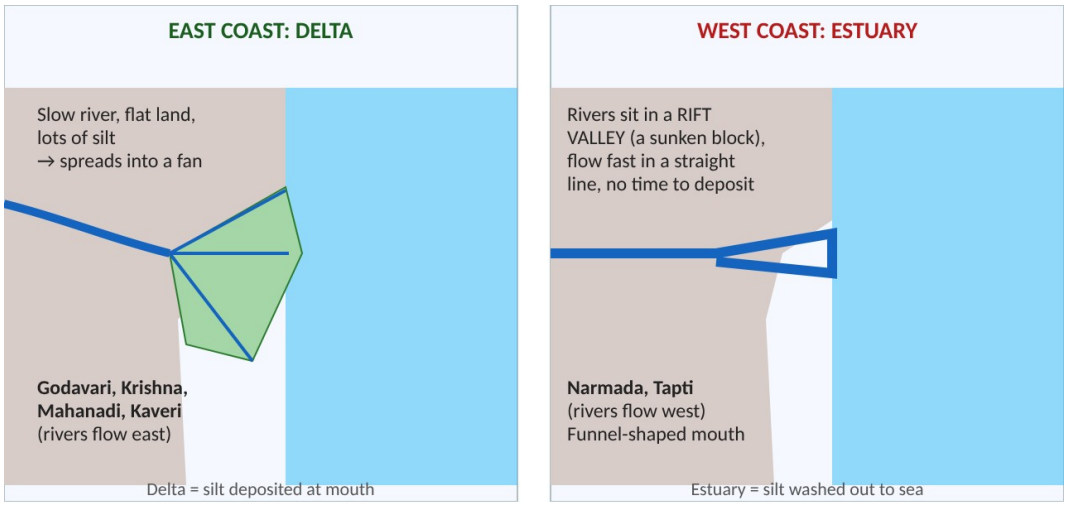


Fig 3.1 Why east-flowing rivers make deltas and west-flowing rivers make estuaries

	Himalayan rivers	Peninsular rivers
Source	Snow and glaciers + rain	Rain only (plateau and Western Ghats)
Flow	All year (perennial)	Seasonal
Course	Gorges, then wandering plains, meanders, shift course	Fixed, shallow, straight
Mouth	Large deltas (Sundarbans)	East: deltas; West: estuaries
Age	Young, still eroding and flooding	Old, stable
Examples	Indus, Ganga, Brahmaputra	Godavari, Krishna, Kaveri, Narmada, Tapti

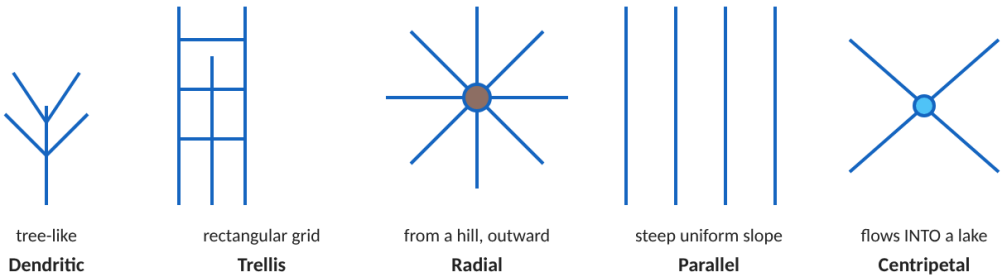


Fig 3.2 Drainage patterns

3.1 Himalayan river systems



Map 3.1 Himalayan rivers (schematic)

Indus system: source near Mansarovar in Tibet; flows through Ladakh to Pakistan. Tributaries in India: **Jhelum, Chenab, Ravi, Beas, Sutlej**. **Indus Waters Treaty (1960; World Bank-brokered): Ravi, Beas, Sutlej** (Eastern rivers) to India; **Indus, Jhelum, Chenab** (Western rivers) to Pakistan. In **April 2025** India put the treaty **'in abeyance'** (current status: verify).
Ganga system: Gangotri glacier (Gaumukh) → **Bhagirathi**; meets **Alaknanda at Devprayag** to become the Ganga; **Haridwar** where it enters the plains; **Prayagraj (Triveni Sangam: Ganga + Yamuna + mythical Saraswati)**; **Varanasi**; **Patna**; **Farakka barrage** (1975, to save Kolkata port); splits into **Bhagirathi-Hooghly** (India) and **Padma** (Bangladesh); with the Brahmaputra it forms the **Sundarbans delta (world's largest delta)**. Length about **2,525 km**. Tributaries: **Yamuna** (from Yamunotri; longest tributary), **Ghaghra, Gandak, Kosi** (the 'Sorrow of Bihar'), **Sone** (from the plateau), **Damodar** ('Sorrow of Bengal'). **Ganga = National River (2008)**; **Namami Gange (2014)** cleaning mission.
Brahmaputra: source in Tibet (**Tsangpo**) → turns around **Namcha Barwa** → enters India as **Siang / Dihang** (Arunachal) → **Brahmaputra** in Assam → **Jamuna** in Bangladesh. **Majuli** (Assam) = world's largest river island. Floods and erosion every year. China's mega-dam plan on the Tsangpo (Medog) is a concern for India (verify details).

3.2 Peninsular river systems (with Andhra Pradesh)



Map 3.2 Peninsular rivers and AP links (schematic)

River	Facts	Dams, projects, links
Godavari	'**Dakshina Ganga**'; longest peninsular river (about 1,465 km); source **Trimbak (Nashik)**; Maharashtra, Telangana, AP, Chhattisgarh; tributaries **Manjira, Pranhita, Indravati**; delta at **Antarvedi**	**Polavaram (Indira Sagar)** national project (AP); **Dowleswaram barrage** (Rajahmundry; Arthur Cotton); **Pattiseema** lift
Krishna	Source **Mahabaleshwar**; Maharashtra, Karnataka, Telangana, AP; tributaries **Tungabhadra, Bhima, Musi**	**Srisailam, Nagarjuna Sagar, Prakasam Barrage (Vijayawada)**; Krishna Water Disputes Tribunal; AP–Telangana water sharing
Tungabhadra	Tunga + Bhadra from the Western Ghats (Karnataka); joins the Krishna near **Kurnool (Sangameswaram)**	**Tungabhadra Dam (Hospet)**; **KC Canal** (Kurnool–Cuddapah), **Sunkesula barrage**, **Handri-Neeva** lift; Kurnool is on its bank
Penna (Pennar)	Source **Nandi Hills (Karnataka)**; flows through Anantapur, Kadapa, Nellore; tributaries **Chitravathi, Kunderu, Papagni**	**Somasila** dam; **Galeru-Nagari**; drought-prone basin
Kaveri	Source **Talakaveri (Kodagu)**; Karnataka and Tamil Nadu; 'Dakshina Ganga' (also used); delta in TN	**KRS, Mettur**; Cauvery dispute (tribunal 2007; Supreme Court 2018)
Mahanadi	Chhattisgarh and Odisha; **Hirakud Dam** (longest earthen dam); Cuttack	Floods in Odisha
Narmada	Source **Amarkantak**; west-flowing in a **rift valley**; estuary at **Gulf of Khambhat**; **Sardar Sarovar Dam**	Narmada Bachao Andolan
Tapti	Source **Multai (MP)**; west-flowing; estuary; **Surat**	—

AP's rivers: Godavari, Krishna, Penna (big) · **Tungabhadra** (Krishna's tributary) · **Vamsadhara, Nagavali, Champavathi, Sarada, Varaha** (north coastal; Eastern Ghats origin) · **Swarnamukhi, Kunderu, Chitravathi, Papagni, Bahuda** (south). Many are **seasonal**, so **tanks, check-dams and lift schemes** are important in Rayalaseema.
Lakes: Kolleru (Eluru/Krishna; large freshwater lake, Ramsar site, bird area), **Pulicat** (AP–TN; India's 2nd largest brackish-water lagoon after **Chilika**), **Wular** (J&K; largest freshwater lake of India), **Dal** (Srinagar), **Loktak** (Manipur; floating 'phumdis'), **Sambhar** (Rajasthan; salt), **Vembanad** (Kerala).

River	Facts	Dams, projects, links

ANDHRA PRADESH / KURNOOL CONNECTION

Kurnool = water-short, river-fed: Tungabhadra (south boundary, Kurnool town on its banks), **Handri and Hundri** (Handri–Neeva Sujala Sravanti brings Krishna water from **Srisailam** to Rayalaseema), **KC Canal**, **Srisailam right canal**, and **Kunderu** (Penna's tributary, Nandyal side). **2024:** the Tungabhadra dam had a crest-gate failure in August (affects Kurnool/Kadapa irrigation). **Vijayawada floods (Sept 2024)** from the **Budameru** and Krishna show flood risk in the delta region.

EXAM TRAPS

Kaveri was called 'Dakshina Ganga' in some books but **Godavari** is the standard answer.
Godavari is the longest PENINSULAR river; the Ganga is the longest in India overall.
Tungabhadra is a tributary of the **Krishna**, not the Penna.
Chilika is the largest; **Pulicat** is the second largest lagoon.

2-MINUTE REVISION

Himalayan: perennial, deltas, floods. Peninsular: seasonal, east → delta; Narmada and Tapti → estuary.
Indus Treaty: Ravi, Beas, Sutlej → India. Ganga: Gangotri → Bhagirathi + Alaknanda (Devprayag).
Godavari longest peninsular; Krishna second. Tungabhadra joins Krishna near Kurnool.
AP lakes: Kolleru, Pulicat.

CHECK YOURSELF

Q1. Meeting of Alaknanda and Bhagirathi: **Devprayag** · Q2. Longest peninsular river: **Godavari** · Q3. Tributary of the Krishna that flows past Kurnool: **Tungabhadra**

CHAPTER 4 — CLIMATE AND MONSOON OF INDIA

Why does India have a 'monsoon' climate?

IN SIMPLE WORDS (what / how / why)

India's climate is called **tropical monsoon**. **Monsoon** = **seasonal reversal of winds**. In summer winds blow from the **sea to land (rain)**; in winter from **land to sea (dry)**.
Why does it happen? (the simple chain) 1. In **summer** the north-west plains heat up and form a **low-pressure** zone. **2.** The **ITCZ** (the belt of low pressure where winds meet) moves north over India. **3.** Air from the **high-pressure Indian Ocean** rushes in, crosses the equator, bends **right** (Coriolis force) and becomes the **south-west monsoon**. **4.** It brings moisture; hitting the Western Ghats and the Himalaya forces it up → **rain**. **5.** In **winter** the land is cold, a **high-pressure** zone forms, and dry winds blow out: the **north-east monsoon**. It picks up moisture over the **Bay of Bengal** and gives rain to the **Tamil Nadu and south coastal AP** coast.
Upper air (jet streams): in winter the **westerly jet** blocks the north; in June it shifts north of the Himalaya and the **tropical easterly jet** helps the monsoon to 'burst'.



Map 4.1 South-west (blue) and north-east (orange) monsoon winds

4.1 Four seasons

Season	Months	What happens
Cold weather (winter)	Dec – Feb	Cold north; **Western Disturbances** bring winter rain to north India (good for **wheat**); **NE monsoon** gives rain to Tamil Nadu and AP coast (Oct-Dec)
Hot weather (summer)	Mar – May	Hot, dry; **Loo** (hot wind), **Kal Baisakhi** (Bengal), **Mango showers** (Kerala/Karnataka). Highest temperatures in May
South-west monsoon	Jun – Sep	Onset in **Kerala about 1 June**; covers India by about **8 July**; 75% of annual rain; **breaks** (dry spells) and **bursts**
Retreating monsoon	Oct – Nov	Winds retreat; **'October heat'**; **cyclones in the Bay of Bengal**; **AP coast gets rain and storms**

Rainfall extremes: Mawsynram (Meghalaya) wettest place on Earth (about 11,000 mm), **Cherrapunji** nearby; **Jaisalmer / Thar** and Ladakh driest. **Rain shadow** of the Western Ghats makes the interior Deccan (**Rayalaseema**) dry.
Factors: latitude, **Himalaya** (blocks cold winds, traps monsoon), **land and sea distribution**, altitude, **El Niño / La Niña** (Vol 1: El Niño → weak monsoon), **Indian Ocean Dipole (IOD)**, and the **jet streams**.
Climate classes (Köppen): Am (monsoon: Kerala, Konkan), **Aw (tropical savanna: most of the Peninsula, including AP)**, **BShw (semi-arid steppe: Rayalaseema, Kurnool)**, BWhw (desert: Thar), Cwg (humid subtropical: Gangetic plain), **Dfc / ET / H** (cold, Himalaya).

ANDHRA PRADESH / KURNOOL CONNECTION

AP's seasons: Summer (Mar-May) very hot (40-45°C in Kurnool, Nandyal, Rayalaseema, and in coastal pockets), **SW monsoon (Jun-Sep)** gives about 60% of rain, **NE monsoon (Oct-Dec)** is **important for the coastal belt and south AP**, **winter (Dec-Feb)** is mild. Average annual rainfall about **900-950 mm** (verify). **Lambasingi** (Alluri Sitarama Raju district) is known as the 'Kashmir of AP' and can see near 0°C winters.
Kurnool / Rayalaseema: semi-arid (500-700 mm), erratic monsoon, droughts every few years, hence **dry farming**, tanks, **Handri-Neeva**, millets, groundnut and cotton. **Heat waves** and **lightning** are regular hazards.

EXAM TRAPS

NE monsoon = retreating winds that pick up moisture in the Bay → rain on **Tamil Nadu/AP coast**, not Kerala.

Season	Months	What happens
Mawsynram wettest; Cherrapunji historically. Monsoon winds **blow from the sea to land in summer**.		

2-MINUTE REVISION

Land heats → low pressure → SW monsoon (Jun-Sep). Winter → NE monsoon (Oct-Dec).
Onset Kerala about 1 June; Western Disturbances help wheat.
AP: BShw in Rayalaseema; coast gets NE monsoon and cyclones.

CHECK YOURSELF

Q1. Monsoon onset in Kerala: **about 1 June** · Q2. Region of AP in rain shadow: **Rayalaseema** · Q3. Wettest place: **Mawsynram**

CHAPTER 5 — SOILS OF INDIA AND ANDHRA PRADESH

What soil is, and which crop grows where

IN SIMPLE WORDS (what / how / why)

Soil = the loose top layer of Earth where plants grow. **How it forms:** rock is slowly **weathered** into small pieces; plants and animals add **humus** (rotted material); water and time mix them. Soil depends on **parent rock, climate, living things, relief and time**.
Soil profile: horizons **A (topsoil), B (subsoil), C (broken rock), R (bedrock)**.

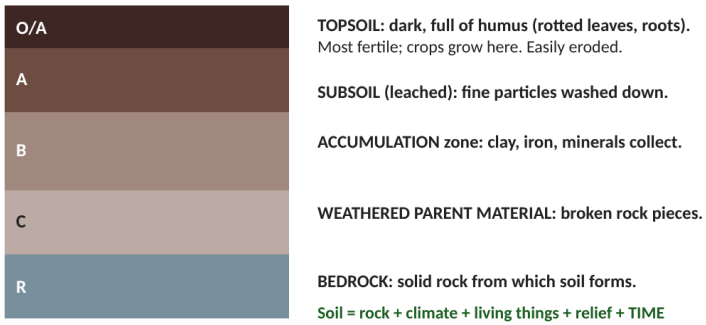


Fig 5.1 Soil profile

Soil (ICAR)	Where	Key features and crops
Alluvial (about 40% of area)	Northern plains, east coast deltas (**Godavari-Krishna**)	Made by rivers; **Khadar** new, **Bhangar** old; rich in potash and lime, **poor in nitrogen and humus**; **rice, wheat, sugarcane, jute**
Black (Regur)	**Deccan lava region**: Maharashtra, MP, Gujarat, Karnataka, **AP (Kurnool, Anantapur, Guntur, Prakasam)**	From basalt; **clay-rich, holds moisture, cracks in summer**, self-ploughing; poor in N, P, humus; **cotton, soybean, pulses, millets, sunflower, groundnut, chillies**
Red and yellow	Tamil Nadu, Karnataka, **AP (Chittoor, Kadapa, Anantapur)**, Odisha, Jharkhand	From granite-gneiss; red from **iron oxide**; porous, low humus; **groundnut, millets, pulses, tobacco, mango**
Laterite	High rainfall with dry spell: Kerala, Karnataka, Odisha, **Eastern Ghats hill areas of AP**	**Leached**; hard when dry; **cashew, coffee, tea, rubber**; bricks
Arid (desert)	Rajasthan, Gujarat	Sandy, saline; low humus; need irrigation (millets, with canals)
Saline / alkaline (Usar, Reh, Kallar)	Dry areas of Punjab-Haryana-UP, **coastal AP**	Salt deposits; poor crops; **gypsum** treats
Peaty / marshy	Kerala (Kuttanad, **Kari**), Sundarbans, Odisha	Rich in organic matter, flooded
Forest / mountain	Himalayan slopes	Vary with altitude; acidic in some areas; tea and apple

ANDHRA PRADESH / KURNOOL CONNECTION

AP soils: red soils are the most widespread; **black soils** cover large parts of **Kurnool, Anantapur, Guntur, Prakasam**; **alluvial** in the **Godavari and Krishna deltas**; **laterite** in the Eastern Ghats; **saline** on the coast. **Kurnool = black cotton soil + red sandy soil (chalka)**, cotton, sunflower, Bengal gram (Kurnool is a big producer), groundnut, paddy under canals (KC canal).
Soil Health Card scheme (2015) tells farmers what their soil needs. **ICAR-NBSS&LUP** in Nagpur does soil survey.

Soil erosion: sheet (thin layer removed), **rill**, **gully** (ravines, e.g. Chambal) and **wind** erosion (desert). **Causes:** deforestation, overgrazing, steep slopes, heavy rain, wrong farming. **Conservation: contour ploughing, terracing, shelter belts, check dams,**

Soil (ICAR)	Where	Key features and crops
afforestation, crop rotation, strip cropping. World Soil Day: 5 December.		

EXAM TRAPS

Black soil = regur = cotton soil but made from **basalt**, not from river silt.
Red soil colour = iron oxide, not richness (it is poor in N, P and humus).
Alluvial soil is the largest by area, **not black soil**.

2-MINUTE REVISION

Alluvial (largest) · Black (cotton; Deccan; Kurnool) · Red (AP, TN) · Laterite · Arid · Saline · Peaty · Forest.
Erosion control: contour, terrace, shelter belt. Soil Health Card 2015.

CHECK YOURSELF

Q1. Which soil is best for cotton: **Black (regur)** · Q2. Most widespread soil in AP: **Red** · Q3. Soil formed by leaching in heavy rain: **Laterite**

CHAPTER 6 — NATURAL VEGETATION, FORESTS AND WILDLIFE

What grows where, and how AP protects it

IN SIMPLE WORDS (what / how / why)

Natural vegetation = plants that grow without human help. It depends mostly on **rainfall, temperature, soil and altitude**: more rain → thicker forest. **Champion and Seth (1968)** classified Indian forests into **16 types** in **5 groups**; here are the simple types:

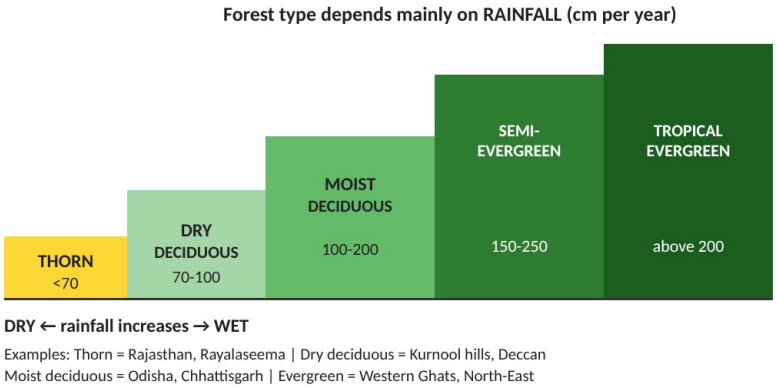


Fig 6.1 Forest types and rainfall

Type	Trees and places
Tropical evergreen	Always green; **Western Ghats, North-East, Andaman**; mahogany, ebony, rosewood
Moist deciduous	Shed leaves in dry season; **Odisha, Chhattisgarh, Western Ghats**; **teak, sal**, sandalwood
Dry deciduous	Most widespread; **Deccan plateau, Kurnool hills, Nallamala**; teak, **Red Sanders**, neem
Tropical thorn and scrub	Dry areas: **Rajasthan, Rayalaseema**; **babul, acacia, khejri, cactus**
Montane	Himalayas: oak, deodar, pine, rhododendron; **alpine** pastures above
Littoral and swamp (mangrove)	Delta coasts: **Sundarbans (largest mangrove, 'Sundari' tree, Bengal tiger)**, **Bhitarkanika (Odisha)**, **Coringa and Krishna mangroves (AP)**

Forest cover: India State of Forest Report (**ISFR**, by **Forest Survey of India, Dehradun**, every 2 years). **ISFR 2023: forest and tree cover about 25.17% of area (verify). National Forest Policy 1988: target 33% (66% in hills). Madhya Pradesh** has the largest forest area; **Mizoram** the highest percentage.
Conservation: National Parks, Wildlife Sanctuaries, Biosphere Reserves, Tiger Reserves (Project Tiger 1973), Project Elephant 1992; **Wildlife Protection Act 1972**; **Forest Rights Act 2006**; **Ramsar sites** (wetlands).

ANDHRA PRADESH / KURNOOL CONNECTION

AP forests: dry deciduous mostly. **Nagarjunasagar-Srisailam Tiger Reserve (NSTR)** (Nallamala hills; Kurnool/Nandyal, Prakasam, Palnadu; **India's largest tiger reserve by area, about 3,700 sq km**, verify). **Seshachalam Hills Biosphere Reserve** (Tirupati/Kadapa; AP's only biosphere reserve, 2010; **Red Sanders**, a rare tree endemic to the Eastern Ghats of AP; smuggling is a big issue).
Kurnool/Nandyal: Rollapadu Wildlife Sanctuary (Great Indian Bustard, **critically endangered**; Rajasthan's state bird) and **Gundla Brahmeswaram**, **Nallamala**; **Rajiv Gandhi Wildlife** areas; **Kolleru** (Ramsar), **Coringa**, **Pulicat bird sanctuary**, **Papikonda National Park** (Godavari), **Sri Venkateswara National Park** (Tirupati), **Sri Lankamalleswara WLS** (Kadapa).

Type	Trees and places

EXAM TRAPS

Sundarbans = largest mangrove; **Coringa** is the AP mangrove.
Great Indian Bustard at **Rollapadu (Kurnool)**.
Red Sanders is in the Seshachalam–Nallamala–Velikonda region (AP), not in the Western Ghats.

2-MINUTE REVISION

Rainfall decides forest type. Dry deciduous = Deccan, Nallamala.
NSTR = largest tiger reserve. Seshachalam = AP's only biosphere reserve. Rollapadu = GIB (Kurnool).
Forest cover about 25%; policy target 33%.

CHECK YOURSELF

Q1. Largest mangrove forest: **Sundarbans** · Q2. GIB sanctuary in AP: **Rollapadu** · Q3. AP's biosphere reserve: **Seshachalam**

CHAPTER 7 — NATURAL HAZARDS AND DISASTER MANAGEMENT

Syllabus: hazards and their management

IN SIMPLE WORDS (what / how / why)

Hazard = a natural event that **can** cause harm (earthquake). **Disaster** = when the hazard hits **people who are vulnerable**.
Disaster risk = Hazard × Vulnerability ÷ Capacity: so we reduce risk by **reducing vulnerability** (strong houses, no building in flood plains) and **increasing capacity** (warnings, trained people).
India's exposure (rough, verify): about **58% of land** is earthquake-prone, about **12%** flood-prone, about **68%** drought-prone, about **5,700 km** of coast cyclone-prone, and the **Himalaya** landslide-prone.

Hazard	Why it happens	Where	What to do
Earthquake	Indian plate pushing Eurasian plate; also inside plate (Latur 1993, Bhuj 2001)	Himalaya, North-East, Kutch, Andaman (**Zones IV–V**)	Earthquake-resistant buildings (BIS codes), awareness, drills
Floods	Heavy rain + river silt + poor drainage; **Brahmaputra, Kosi, Ganga**	Bihar, Assam, UP, **Godavari and Krishna deltas**	Embankments, **early warning**, no building on flood plains, **dam management**
Cyclones	Warm Bay of Bengal sea (Vol 1)	**Odisha, AP, TN, WB** coasts (Oct-Nov)	**IMD warnings**, cyclone shelters, evacuation, **mangroves** as shields
Drought	Weak monsoon, El Niño, over-use of groundwater	**Rayalaseema**, Marathwada, Rajasthan, Bundelkhand	**Watershed development, drip irrigation**, drought-resistant crops, **tanks**
Landslides, avalanches	Steep slopes + heavy rain + deforestation + roads	Himalaya, **Western Ghats**, Nilgiris	Slope stabilisation, no building on unstable slopes
Tsunami	Undersea earthquake (2004 Indian Ocean)	East coast, Andaman	**INCOIS Hyderabad** warning (ITEWC), coastal buffer
Heat wave, cold wave, lightning	Dry air, high pressure	Telangana, **Rayalaseema, coastal AP**, plains	Heat action plans, lightning alerts, awareness

Institutions: Disaster Management Act 2005 (after the Dec 2004 tsunami) → **NDMA** (chair: Prime Minister), **NDRF** (response force), **SDMA** (chair: Chief Minister) and **DDMA** (chair: District Collector). **SDRF** at state level.
Global: Sendai Framework for Disaster Risk Reduction 2015-2030. **India: Coalition for Disaster Resilient Infrastructure (CDRI)**, launched by India in 2019.
Four stages of DM: Mitigation → Preparedness → Response → Recovery. Early warning (IMD for cyclones and rain, INCOIS for tsunamis, CWC for floods).

ANDHRA PRADESH / KURNOOL CONNECTION

APSDMA (AP State Disaster Management Authority) handles warnings (SMS, lightning, cyclone). **Cyclones:** Hudhud (2014, Visakhapatnam), Titli (2018), **Michaung (Dec 2023)**, Montha (Oct 2025). **Floods:** Godavari (annual), **Krishna and Budameru (Vijayawada, Sept 2024)**. **Drought: Kurnool, Anantapur, Kadapa** frequent. **Lightning** deaths in Rayalaseema and coastal districts. **Mitigation:** cyclone shelters, **mangrove belts (Coringa)**, tank restoration, watershed work.

EXAM TRAPS

Hazard ≠ disaster. NDMA chair = **Prime Minister**; SDMA = **Chief Minister**; DDMA = **District Collector**.
DM Act 2005, not 2000. Sendai = **2015-2030**.
Tsunami warning: **INCOIS**; cyclone warning: **IMD**.

2-MINUTE REVISION

Risk = hazard × vulnerability ÷ capacity. DM Act 2005; NDMA; NDRF; Sendai.
Mitigation → Preparedness → Response → Recovery.
AP: cyclones (Oct-Nov), Rayalaseema drought, delta floods, lightning.

CHECK YOURSELF

Q1. Disaster Management Act: **2005** · Q2. Head of SDMA: **Chief Minister** · Q3. Tsunami early warning body: **INCOIS**

CHAPTER ★ — FINAL SELF-TEST: 30 QUESTIONS

Answers follow the table. Cover the answer column first.

#	Question	Answer
1	Standard meridian of India:	**82°30′E**
2	Longest land border is with:	**Bangladesh**
3	India's highest peak:	**Kanchenjunga (8,586 m)**
4	Highest peak entirely in India:	**Nanda Devi**
5	Outermost Himalayan range:	**Shiwaliks**
6	Belt of pebbles at foothills:	**Bhabar**
7	Belt of swampy land after Bhabar:	**Terai**
8	Oldest fold mountains of India:	**Aravalli**
9	Highest peak of Western Ghats / South India:	**Anamudi (2,695 m)**
10	Highest peak of the Eastern Ghats (AP):	**Jindhagada / Arma Konda (about 1,680 m)**
11	Only active volcano of India:	**Barren Island**
12	Channel between Andaman and Nicobar:	**Ten Degree Channel**
13	Coastline between Mahanadi and Krishna:	**Northern Circars**
14	Longest peninsular river:	**Godavari**
15	River that joins the Krishna near Kurnool:	**Tungabhadra**
16	Rivers that flow west in rift valleys:	**Narmada and Tapti**
17	India gets which Indus tributaries under the 1960 Treaty:	**Ravi, Beas, Sutlej**
18	Island in the Brahmaputra, world's largest river island:	**Majuli**
19	Pulicat is India's ___ largest lagoon:	**Second (after Chilika)**
20	Largest freshwater lake of India:	**Wular**
21	Monsoon onset in Kerala:	**About 1 June**
22	Winter rain in north India from:	**Western Disturbances**
23	Wettest place on Earth:	**Mawsynram**
24	Region of AP in the rain shadow:	**Rayalaseema**
25	Soil for cotton (regur) comes from:	**Basalt lava (Deccan Traps)**
26	Soil Health Card scheme started:	**2015**
27	Largest mangrove forest:	**Sundarbans**
28	Great Indian Bustard sanctuary in AP:	**Rollapadu (Kurnool)**
29	AP's only biosphere reserve:	**Seshachalam Hills**
30	Chair of NDMA:	**Prime Minister (DM Act 2005)**

NEXT STEPS

Scoring: 25+ strong · 20-24 good (re-read green boxes) · below 20 (re-read blue boxes of weak chapters). Send me the numbers you got wrong and I will make a short revision set.
Redraw challenge: on a blank India outline, mark from memory: Himalaya, Aravalli, Western and Eastern Ghats, Ganga, Brahmaputra, Godavari, Krishna, Tungabhadra, Kurnool. Check with Maps 2.1, 3.1, 3.2.
Next volume: Geography Vol 3 — Andhra Pradesh and India: Resources, Agriculture, Industries, Transport (economic and human geography).